

JSGP · SHINE HACKATHON

# Morning Triage

The morning triage for One Care caseworkers —  
who needs you first, and why, in plain language.

## THE PROBLEM

# Help that must be **asked for**

Alarm buttons and pendants only help while the fallen person can still press them. An unconscious senior presses nothing.

The fall becomes a **long lie** — hours helpless on the floor before anyone thinks to check.

## THE INSIGHT

# The home **already** knows

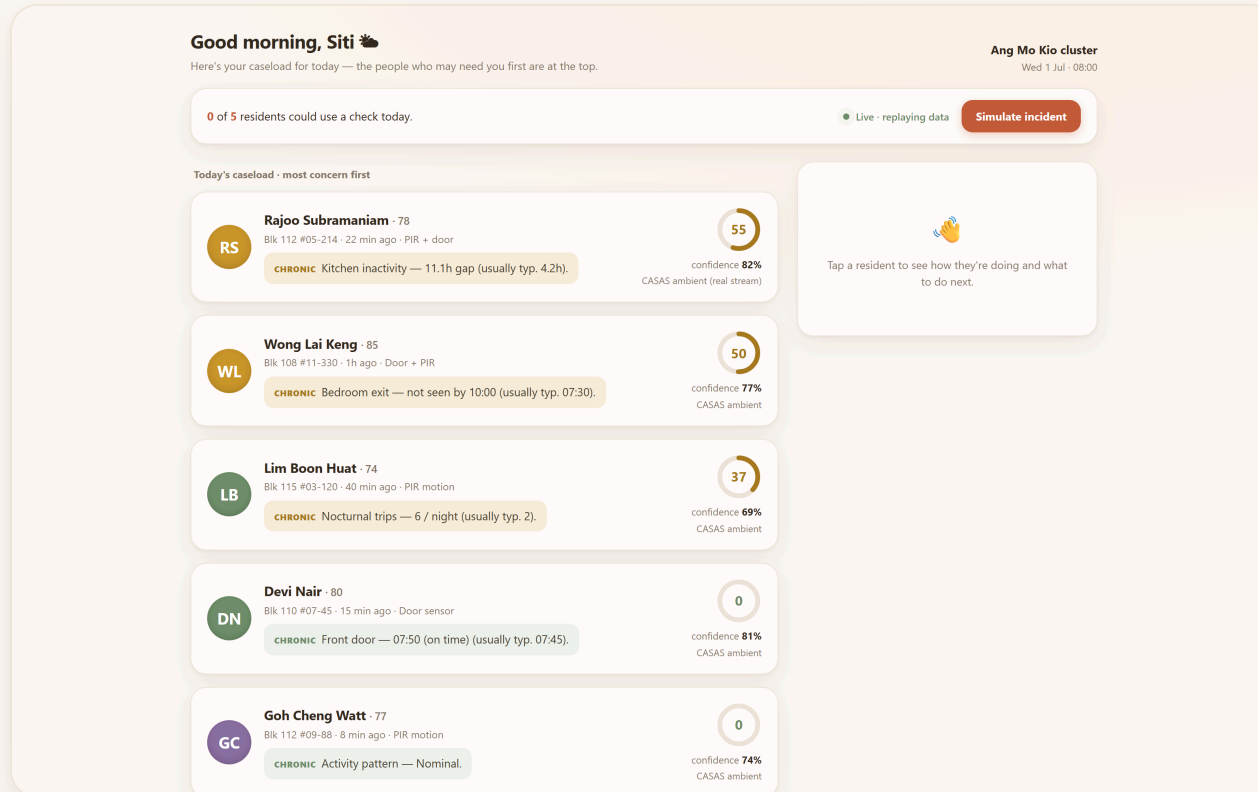
Routine is a vital sign, and absence of motion is data.

The ambient track asks the senior to wear nothing, press nothing, and puts no camera in the room.

One optional bangle adds a second layer —  
falls caught in seconds instead of hours.

## THE PRODUCT

# One screen, calm by default



Every resident is ranked by need, and every row explains itself in one sentence.

## THE DEMO BEAT

# A fall preempts everything

0 of 5 residents could use a check today.

● Live · replaying data

Simulate incident

The button replays a real recorded fall through the live detector — the same signal a worn bangle would stream

FALL DETECTED

TA

**Tan Ah Moi** · 82  
Blk 108 #04-210 · just now · Accelerometer

**ACUTE** Fall detected — 11.2 g impact then 12 s no movement.

Call now

Escalate

100

confidence **100%**  
Wearable bangle

## WHAT TO DO NEXT

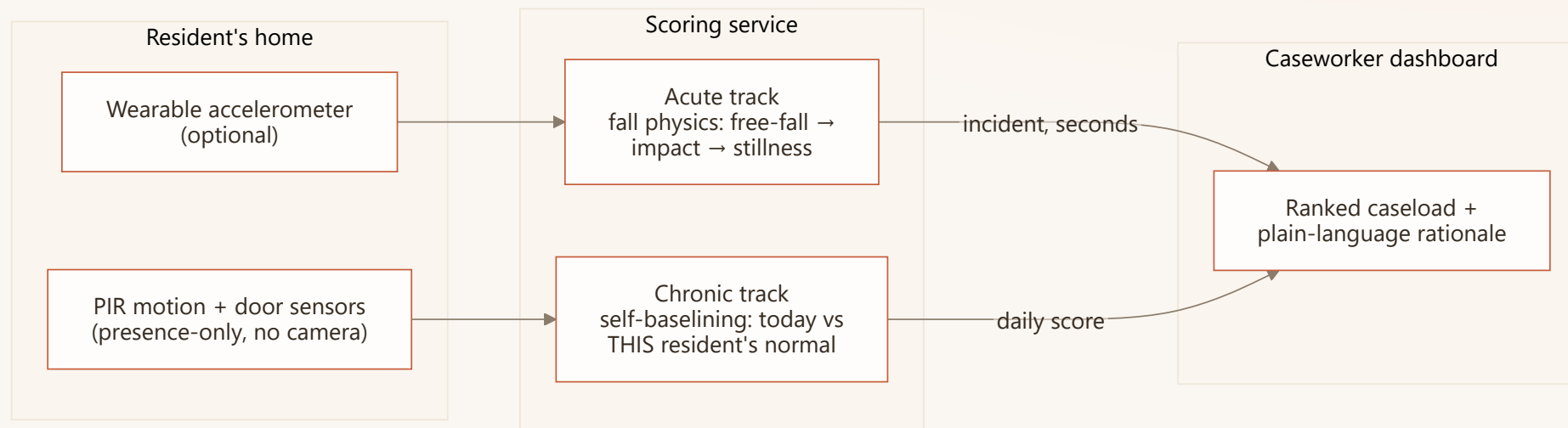
**CALL NOW.** If no response, escalate to MOH 24/7 line + dispatch.

The fall pins to the top in seconds and states its evidence in plain words

The system suggests the next step. The caseworker makes the call

## HOW IT WORKS

# Sense → Score → Decide



A fall must show an **ordered signature** — a free-fall dip, an impact spike within half a second, then stillness. A dropped phone cannot fake the sequence, and every score comes from deterministic rules a caseworker can read.

## EXPLAINABLE BY CONTRACT

# Every number **decomposes**

### WHY SHE'S RANKED HERE

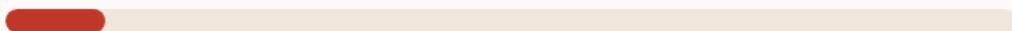
Peak impact **11.2 g**  
walking < 1.4 g



Post-impact stillness **12 s**



Orientation **Horizontal**



Every score breaks into the sensor facts that produced it —  
nothing is taken on faith

**100**  
ACUTE RISK

**100%**  
CONFIDENCE

**Accelerometer**  
SOURCE

A stale sensor lowers confidence. It never inflates risk, so a dead  
sensor cannot fake a crisis

The system ranks and explains.

**A human decides.**

VALIDATED ON REAL DATA

# It's **real**, not a concept

**96.2%**

OF 1,798 REAL FALLS DETECTED · SISFALL

**29.8% → 0–5%**

LAB ALARMS → ELDERLY-TYPICAL MOVEMENT

**220 days**

ONE REAL HOME · CASAS ARUBA

Nothing is trained. The thresholds were calibrated openly on the full SisFall set, and the lab false alarms concentrate in jogging and jumping — movements a monitored senior rarely performs.



**Rajoo Subramaniam** · 78

Blk 112 #05-214 · 22 min ago · PIR + door

**CHRONIC** Kitchen inactivity — 11.1h gap (usually typ. 4.2h).



confidence **82%**  
CASAS ambient (real stream)

Rank 1 carries 220 real days of one elderly resident — the system found the broken day on its own



BUILT FOR SINGAPORE

## No imported prior

- **Self-baselining.** The system measures each resident's own kitchen gaps, door times and night trips for about two weeks, then flags deviation from that person's own normal.
- **Presence-only dignity.** Seniors accept what does not watch them.
- **Per-HDB deployment.** An installer fills in a one-page sensor map. The product multiplies One Care's existing people and replaces no one.

## HONEST LIMITS

# What we **don't** claim

- It shrinks time to detection. It does not promise to catch every event.
- Fall thresholds were calibrated on mostly young-adult recordings, so the senior false-alarm rate is extrapolated rather than measured.
- A missed fall — or a bangle left on the nightstand — degrades to the ambient track, where hours of stillness still surface that resident by morning.
- An irregular life earns wide baselines. The system says so through low confidence instead of hiding it.

## BEYOND THE POC

# From demo to **pilot**

- **Escalation that completes the loop.** Automated welfare calls, and a handoff to the MOH/AIC line when a fall goes unacknowledged.
- **A morning brief per flagged resident.** Auto-drafted wording, grounded strictly in the deterministic features. The drafting layer never touches a score.
- **One One Care cluster to start.** Measure time to detection and caseworker minutes saved per shift.

## THE ASK

# Pilot **with us**

One cluster. Commodity sensors. A working system —  
validated on **4,505 real recordings** and a **220-day real home**.

Built at SHINE Hackathon. The demo runs live, replaying real recordings.

## Why no machine learning

- **No Singapore training data exists.** A model trained on Colombian or American recordings would import exactly the prior we refuse to assume.
- **A fall is physics.** The free-fall, impact and stillness sequence transfers across bodies and homes without any training step.
- **A caseworker must be able to ask why.** Decomposable rules survive that question. A black box does not.
- Where a model helps later — smoothing the wording of the morning brief. It will never produce a score.

## Where the **data** goes

| DATASET                                                                                    | ROLE IN THE SYSTEM                                                                      |
|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>SisFall</b> · Universidad de Antioquia · 4,505 recordings, 1,798 real falls             | Calibrates the fall thresholds, measures the detector, and supplies the live demo trace |
| <b>CASAS Aruba</b> · Washington State University · 220 days of one elderly resident's home | Validates that self-baselining recovers a real routine — and finds the day it broke     |

Nothing is trained on either dataset. Both are provenance-pinned with checksums in a committed lock file, and every number reproduces from one script.

BACKUP · FOR Q&A

## What a flat **needs**

- Motion sensors in the rooms that carry routine — kitchen, bedroom, bathroom, living room.
- One contact sensor on the front door.
- One optional bangle for the acute track. The routine track works with nothing worn at all.
- An installer fills in a one-page sensor-to-area map. No cameras, no microphones, no rewiring.